Generalized Linear Models

Introduction and Overview

Prof. Dr. Andreas Groll & Guillermo Briseno Sánchez

Winter term 20/21

Introduction and Overview - Content

Organisatorisches

Inhalte

Organizational stuff

Lecture

- Wednesday, 12-14, temporarily on Zoom (M/E 21)
- Thursday, 10-12 Uhr, temporarily on Zoom (CDI 120)

Andreas Groll (groll@statistik.tu-dortmund.de)

Excercise

- Monday, 16-18, Zoom

Guillermo Briseno Sánchez (briseno@statistik.tu-dortmund.de)

Organizational stuff

Website

https://moodle.tu-dortmund.de/course/view.php?id=21765

Enrollment key: GLMs20/21

Exams

- 1st date: probably end of January 2021
- 2nd date: probably end of March/beginning of April 2021

Goals of this course

Get familiar with advanced regression techniques

- Generalized regression models to analyse panel data
- Categorical regression
- Variable selection via penalization
- How to deal with longitudinal data

Learning objective: Acquire competences ...

... to choose, compare and interprete suitable models for a given data situation.

Content

- Based on the course from Annika Hoyer (and colleagues, LMU Munich)
- No complete manuscript, but:

$\Longrightarrow$ lecture: a mixture out of slides and hand-written parts

Literature

- Fahrmeir, Kneib, Lang, Marx (2013): Regression - Models, Methods and Applications. Springer, Berlin, Heidelberg.
- Fahrmeir, Tutz (2013): Multivariate statistical modelling based on generalized linear models. Springer Science & Business Media
- Tutz (2011): Regression for categorical data. Cambridge University Press

Course overview

1. Basic concepts of regression

Course overview

$$Y \sim \mathcal{N}(\mu, \sigma^2)$$

$\not\uparrow$

1. Basic concepts of regression
2. Binary regression $\longrightarrow \quad Y \in \{0, 1\}$

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models
4. Multi-categorical regression $Y \in \{$ Dem, Rep, Conn., ... $\}$

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models
4. Multi-categorical regression
5. Semi- and non-parametric regression

Course overview

1. Basic concepts of regression
2. Binary regression
3. Generalized linear models
4. Multi-categorical regression
5. Semi- and non-parametric regression
6. Multivariate/ extended regression models